

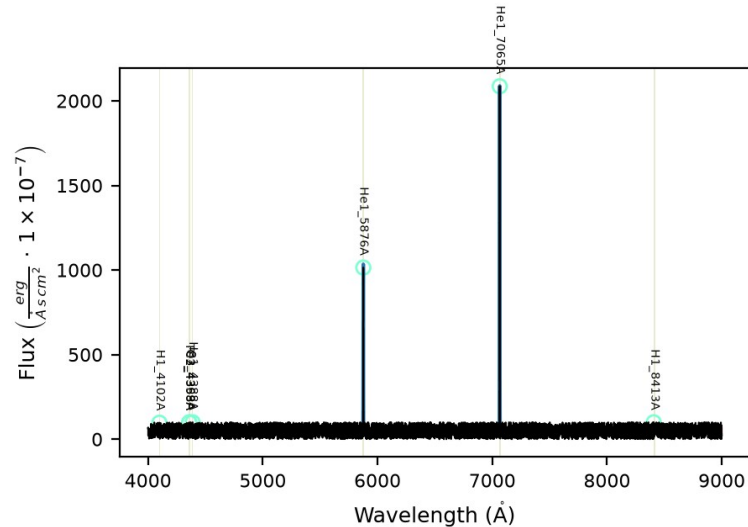
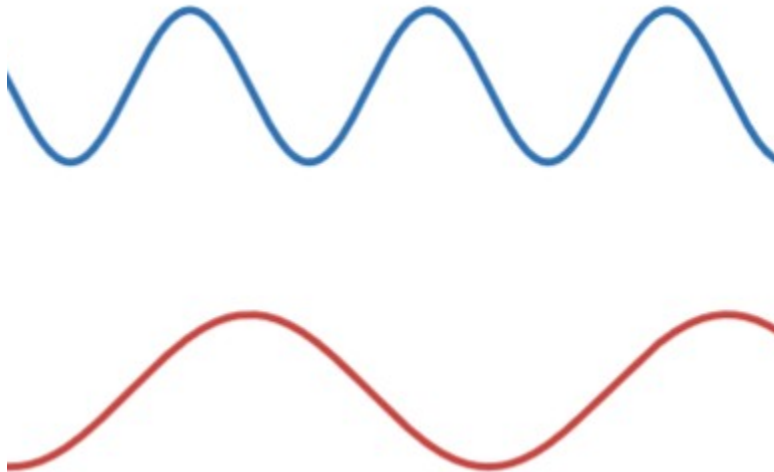
Prediction of Little Red Dot Density Through JWST Spectroscopy

By Ian Bishop, Akshaj Vyas, and
Kelcey Davis.



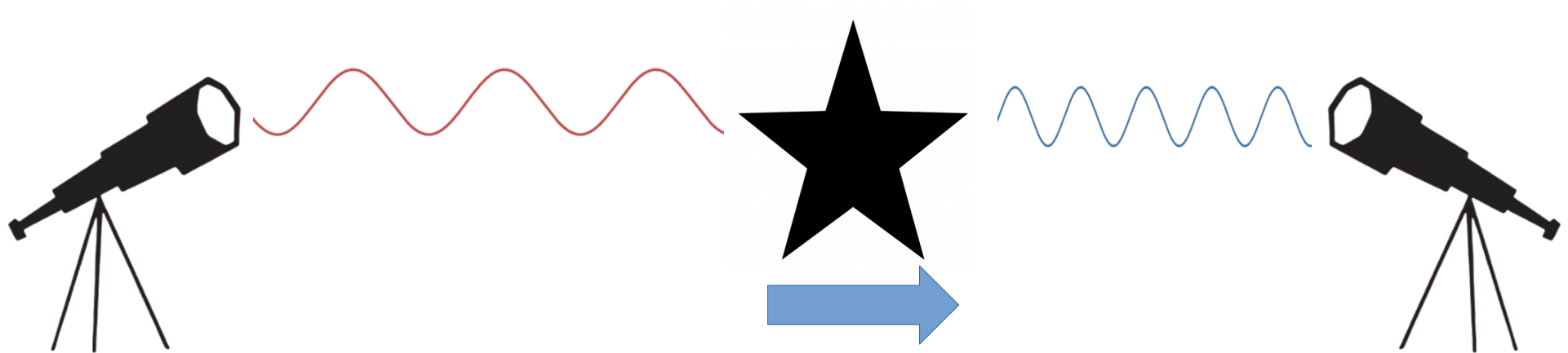
Introduction to Spectroscopy

- **Wavelength:** the distance between peak to peak of a light wave
- **Flux:** the amount of light emitted per wavelength



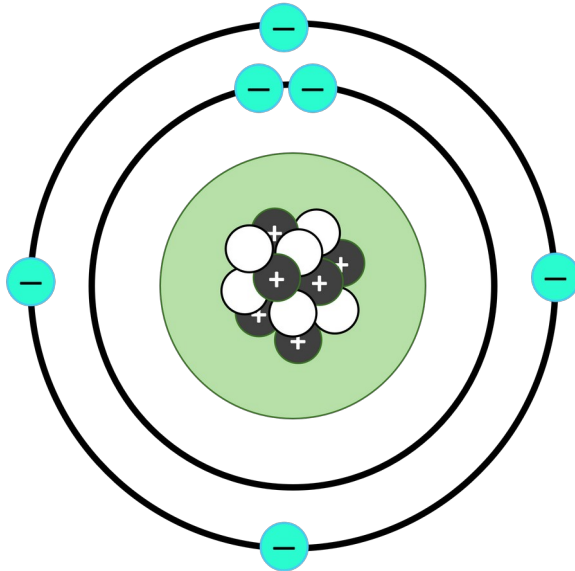
Scientific Background – Doppler effect

- Doppler effect
- Redshift vs Blueshift



Scientific Background – Excited Electrons

- Electrons absorb energy and jump to the next level
- Fall back down and emit specific wavelengths



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Term Explanation

TERM	EXPLANATION
LRD	Little Red Dot
SMBH	Supermassive Black Hole
SMC	Small Magellanic Cloud
JWST	James Webb Space Telescope

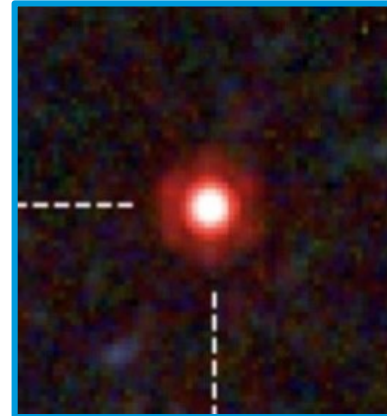
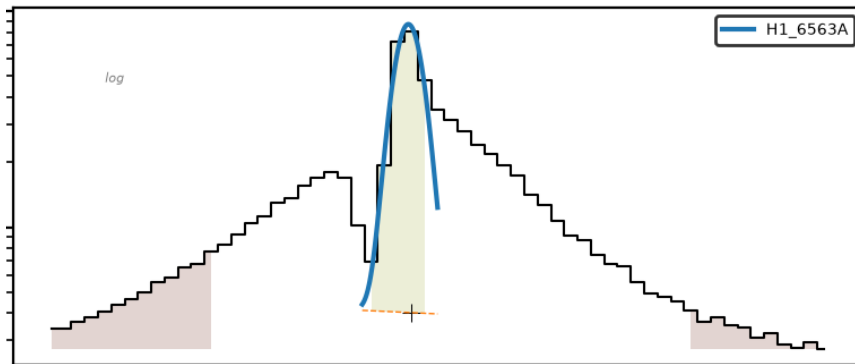


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History

- Discovered by JWST images
- Originally considered to be early galaxies
- Hypothesized “cocoon” of gas surrounding dense SMBHs

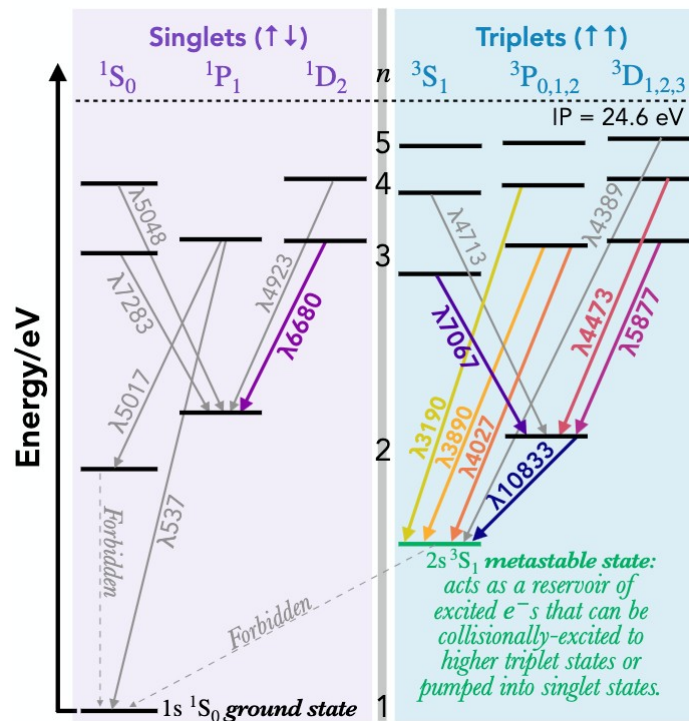
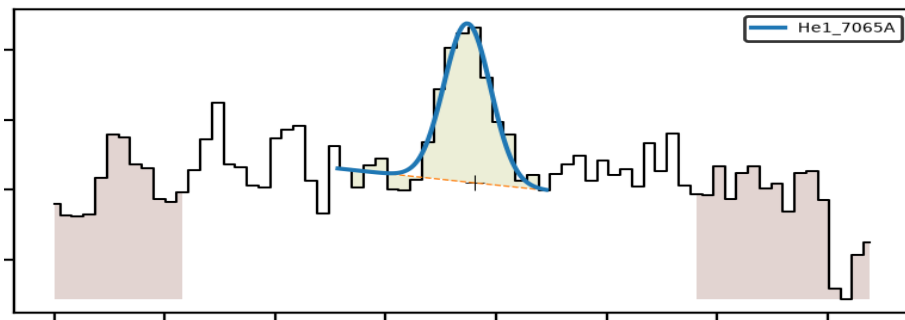


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Helium

- Two line ratios will be measured
 - He I 7065Å / He I 5876Å
 - He I 10830Å / H I Pay
- “Measuring” means integrating around the wavelength



Danielle A. Berg et al 2026 ApJ 996 68

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Goals

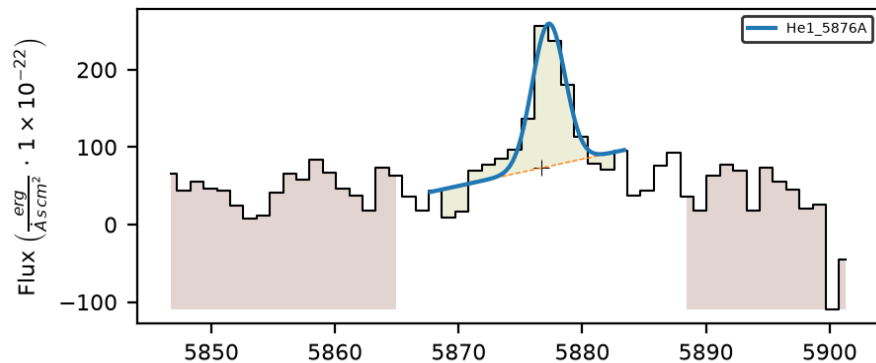
- **Predict density of LRDs using Helium emission lines**
- **Determine the accuracy of generated predictions**
- **Build a foundation for more accurate future simulations**

Motivation

- To understand the early universe better
- To come up with a conclusive theory for LRDs
- To determine how SMBHs could have formed

Flux Measurement

- Use a Gaussian distribution to accurately model flux
- Calculate ratios between paired fluxes



LRD	He I 5876Å flux	He I 7065Å flux
OCEANS 20504 P4	3.67 ± 0.72	8.49 ± 0.10
OCEANS 20504 P1	8.51 ± 1.14	4.79 ± 0.95
OCEANS 169045 P4	7.52 ± 0.68	3.84 ± 0.57
OCEANS 102364 P5	2.98 ± 0.83	2.96 ± 0.53

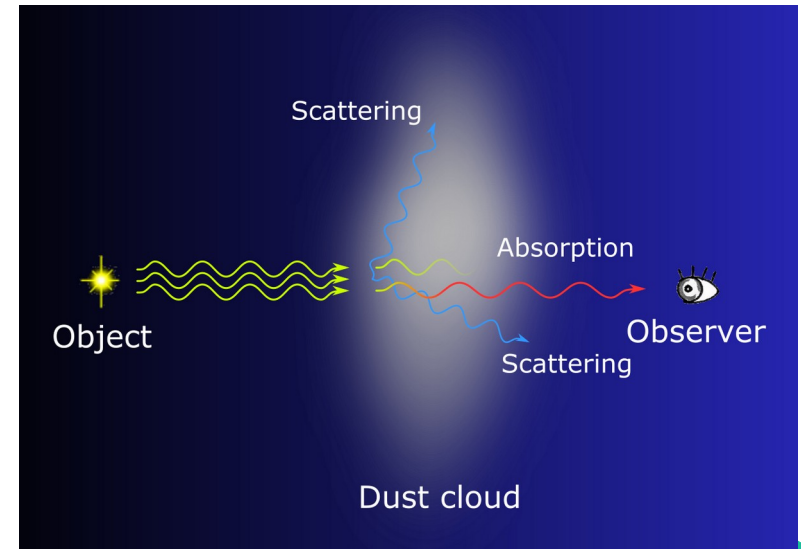
LRD	He I 10830Å flux	$Pa\gamma$ flux
JADES 28074 G235M	$82.11^{+1.303}_{-1.400}$	22.1 ± 1.27
JADES 28074 G395M	$82.11^{+1.303}_{-1.400}$	21.2 ± 1.69
RUBIES 40579 G395M	70^{+4}_{-3}	40.5 ± 4.23
RUBIES 154183 G395M	$2.7^{+5.8}_{-5.8}$	2.74 ± 1.44
GLIMPSE 17775 G395M	$24.58^{+2.334}_{-2.334}$	5.16 ± 0.29

Dust Correction

- Dust clouds interrupt light's travel
- Causes bluer fluxes to decrease
- Correct by modeling a dust extinction curve (G24_SMC Avg)

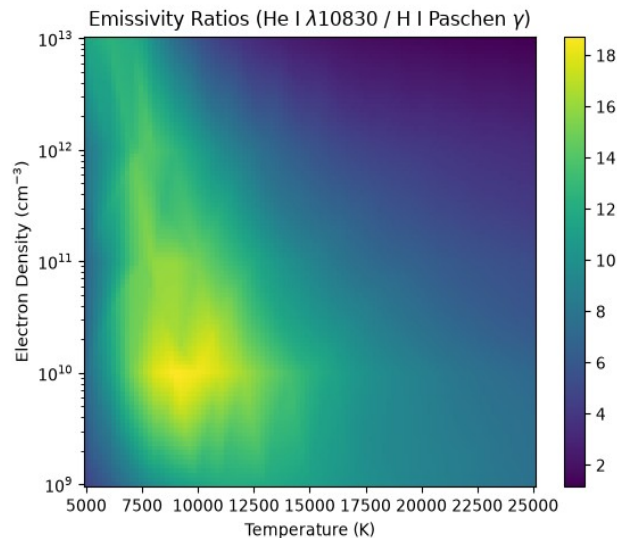
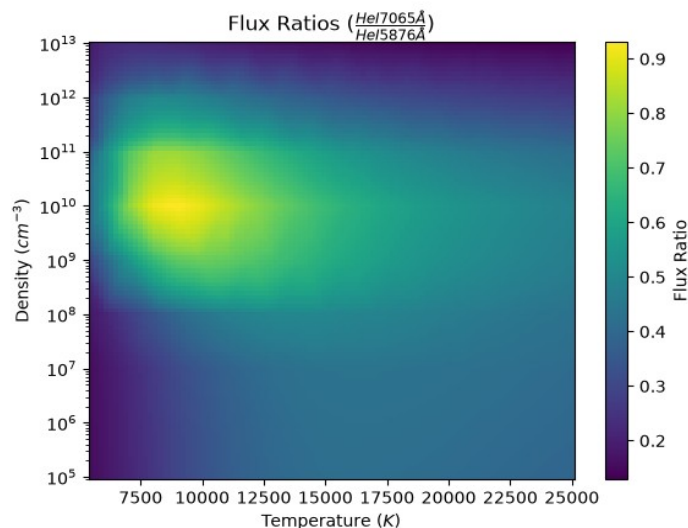
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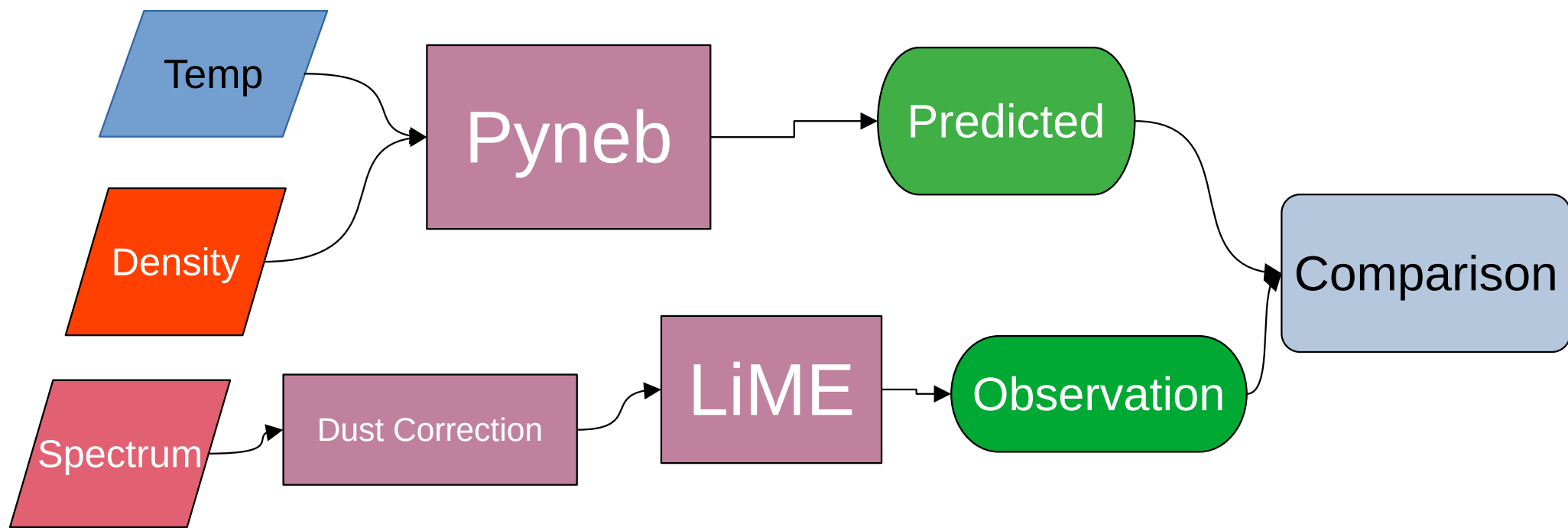


Theoretical Predictions

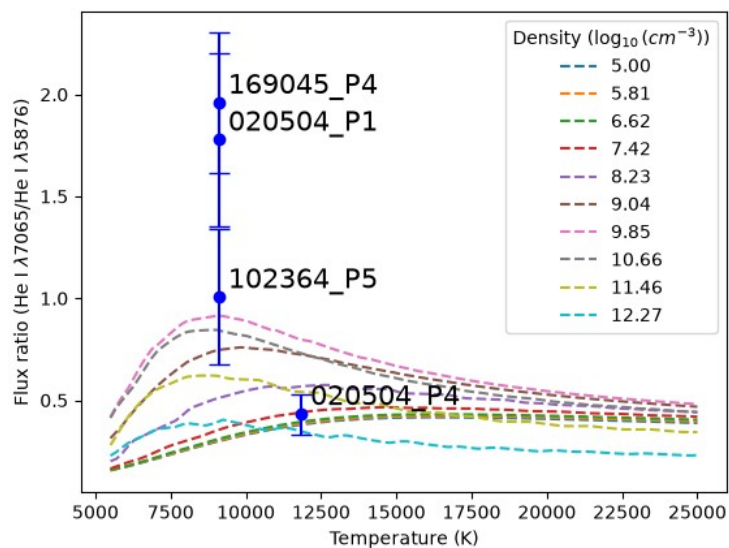
- Generate a wide array of theoretical conditions
- Run simulation on each combination
- Compare theoretical combinations with observational values



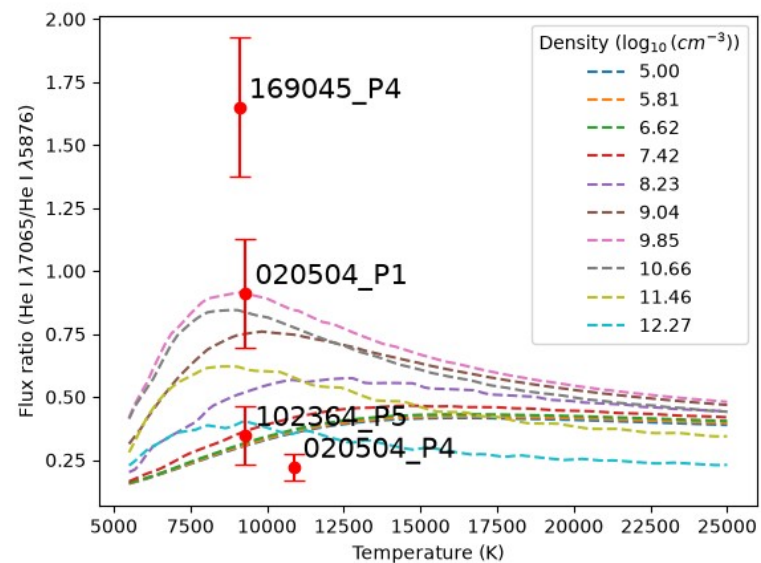
Summary of Methods



Comparison 1

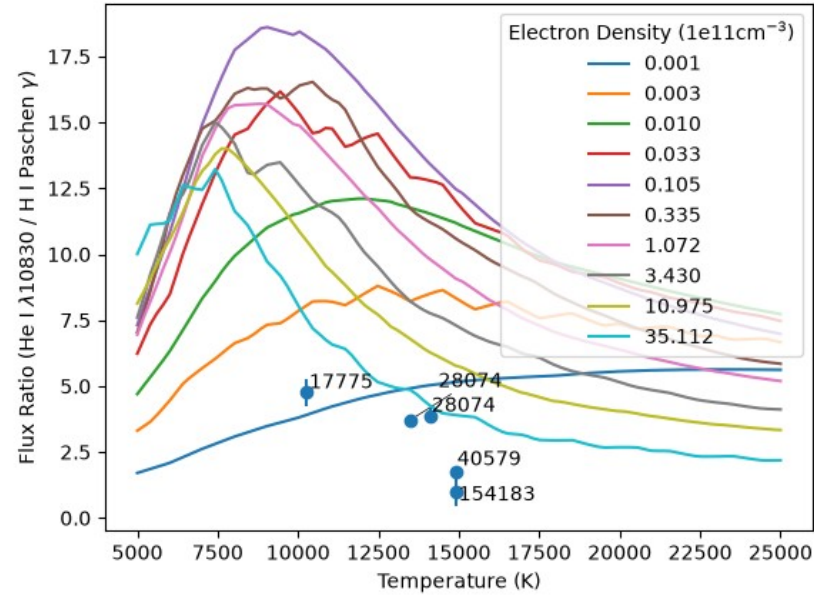


Without Dust Correction



Dust Correction

Comparison 2



Discussion – Predicted Physical Conditions

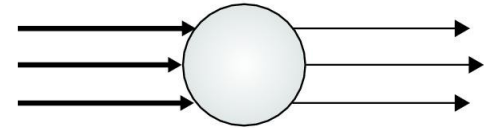
- Flux values exceed Pyneb's predictions
- Density $> 10^{10} \text{ cm}^{-3}$, temperature $\sim 10000 \text{ K}$.
- High flux ratios support enshrouded gas hypothesis!

Discussion – Limitations

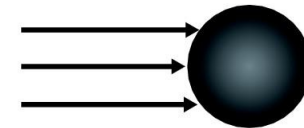
- **High density indicates optical thickness; Pyneb assumes optically thin region**
- **High flux = Collisional effects; Pyneb assumes recombination**
- **Future work: use CLOUDY to resolve these issues**

Akimasa Kataoka, Satoshi Okuzumi,
Hidekazu Tanaka and Hideko
NomuraA&A, 568 (2014) A42

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optically thin



optically thick



Discussion – Dust Correction

- Dust correction reduces observed flux values
- Some exceed Pyneb predictions despite correction
- Cause behind high flux: Pyneb or dust clouds? Ian's data indicates Pyneb is the root cause.

Thank You

Questions?

Key Citations

These are key citations that we mention while presenting. For a full bibliography containing all of the information used in this report, please see the corresponding report to this presentation.

Kocevski, D. D., Finkelstein, S. L., Barro, G., et al. 2025, The Rise of Faint, Red Active Galactic Nuclei at $z > 4$: A Sample of Little Red Dots in the JWST Extragalactic Legacy Fields, *The Astrophysical Journal*, 986, 126, doi: [10.3847/1538-4357/adbc7d](https://doi.org/10.3847/1538-4357/adbc7d)

Labbé, I., van Dokkum, P., Nelson, E., et al. 2023, A population of red candidate massive galaxies ~ 600 Myr after the Big Bang, *Nature*, 616, 266, doi: [10.1038/s41586-023-05786-2](https://doi.org/10.1038/s41586-023-05786-2)

Gordon, K. D., Fitzpatrick, E. L., Massa, D., et al. 2024, Expanded Sample of Small Magellanic Cloud Ultraviolet Dust Extinction Curves: Correlations between the 2175 Å Bump, q_{PAH} , Ultraviolet Extinction Shape, and $N(\text{H I})/A(V)$, , 970, 51, doi: [10.3847/1538-4357/ad4be1](https://doi.org/10.3847/1538-4357/ad4be1)